

Math 381 Fourier Analysis and Boundary Value Problems for ISP Notes

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1 Partial Differential Equations

Example 1.1. Find the function $u(x, y)$ such that $u : \mathbb{R}^2 \mapsto \mathbb{R}$ that satisfies the differential equation $u_{xy} = 0$

We can just integrate twice. Once with respect to x and one with respect to y .

$$\begin{aligned}u_{xy} &= 0 \\ \Rightarrow u_x &= c(x) \\ \Rightarrow u(x, y) &= C(x) + D(y)\end{aligned}$$

Example 1.2. Find a solution $u(x, y)$ to solve the transport equation

$$u_x + yu_y = 0$$

Conceptually we can think of this as $\vec{v} \cdot \nabla u = D_{\vec{v}}u = 0$ where $\vec{v} = \langle 1, y \rangle$. We effectively want to travel along a path such that the directional derivative in $\vec{v}$ is 0. These paths are called integral curves as they are tangent to the vector field at every point. We now arrive at the characteristic equation

$$\frac{dy}{dx} = \frac{y}{1}$$

since the slope at every point is just rise over run and therefore $\frac{y}{1}$. This leads to the solution

$$y(x) = Ae^x$$

where A is found using an initial condition. We note that since the value of the function doesn't depend on where you are on each curve but rather *which* curve you're on, we can write $u(x, y) = f(A)$ where $A = ye^{-x}$. Therefore the general solution to the transport equation is

$$u(x, y) = f(ye^{-x})$$

1.1 Initial and Boundary Conditions

When we have an n -order derivative with respect to one variable in an equation, we then need n conditions in order to uniquely determine a solution to a boundary value problem. For derivatives with respect to time, these are generally deemed initial conditions because they occur at $t = 0$. For variables associated with some sort of domains, these are usually deemed boundary conditions since they occur at the boundary of the system of interest. It is worth noting that some boundary conditions can lead to nonsensical and impossible implications. Let us take the boundary value problem

$$y'' + y = x \quad y(0) = y(\pi) = 0$$

The solution to the homogeneous equation is $A \cos(x) + B \sin(x)$ and the solution to the particular is x . Therefore the general solution is $y_g(x) = A \cos x + B \sin x + x$. Plugging in the boundary values yields the following

$$\begin{aligned}y(0) &= 0 = A \\ y(\pi) &= 0 = \pi\end{aligned}$$

For PDEs we generally do not have to worry about dimensions of boundary conditions because generally it's a boundary in space and is "infinite dimensional."

Theorem 1.3. *Every solution to the wave equation can be written in the form*

$$u(t, x) = p(\xi) + q(\eta) = p(x - ct) + q(x + ct)$$

Proof. We change our variables to get

$$u(t, x) = v(x - ct, x + ct) = v(\xi, \eta) = u\left(\frac{\eta - \xi}{2c}, \frac{\eta + \xi}{2}\right)$$

By the chain rule, we get

$$\frac{\partial u}{\partial t} = c\left(-\frac{\partial v}{\partial \xi} + \frac{\partial v}{\partial \eta}\right) \quad \frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 v}{\partial \xi^2} - 2\frac{\partial^2 v}{\partial \xi \partial \eta} + \frac{\partial^2 v}{\partial \eta^2}\right) \quad (1)$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial \xi} + \frac{\partial v}{\partial \eta} \quad \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial \xi^2} + 2\frac{\partial^2 v}{\partial \xi \partial \eta} + \frac{\partial^2 v}{\partial \eta^2} \quad (2)$$

This implies

$$\square u = -4c^2 \frac{\partial^2 v}{\partial \xi \partial \eta} = 0 \text{ iff } \frac{\partial^2 v}{\partial \xi \partial \eta} = 0 \quad (3)$$

If we create a new variable $w = \frac{\partial v}{\partial \eta}$, we get from the condition in (3) that $\frac{\partial w}{\partial \xi} = 0 \therefore w = r(\eta)$. We then get the differential equation

$$\frac{\partial v}{\partial \eta} = r(\eta) \rightarrow \int \frac{\partial v}{\partial \eta} d\eta = \int r(\eta) d\eta$$

By performing the integration with respect to η , we then get a constant as a function of ξ which leads to the following

$$v(\xi, \eta) - p(\xi) = q(\eta) \rightarrow v(\xi, \eta) = p(\xi) + q(\eta)$$

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